

**E****COMPOUND ANGLE IDENTITIES**

$$\begin{aligned}\cos(A \pm B) &= \cos A \cos B \mp \sin A \sin B \\ \sin(A \pm B) &= \sin A \cos B \pm \cos A \sin B \\ \tan(A \pm B) &= \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}\end{aligned}$$

(3.10)

EXERCISE 1E

- 1 Use your calculator to verify all six of the compound angle identities for $A = 57^\circ$ and $B = 21^\circ$.

2 Expand and simplify:

1E a $\sin(90^\circ + \theta)$

b $\cos(90^\circ + \theta)$

c $\sin(180^\circ - \theta)$

d $\cos(\pi + \alpha)$

e $\sin(2\pi - A)$

f $\cos(\frac{3\pi}{2} - \theta)$

g $\tan(\frac{\pi}{4} + \theta)$

h $\tan(\theta - \frac{3\pi}{4})$

i $\tan(\pi + \theta)$

$$(a) \sin(90^\circ + \theta) = \sin 90^\circ \cos \theta + \cos 90^\circ \sin \theta$$

$$= (1) \cos \theta + (0) \sin \theta$$

$$\boxed{\sin(90^\circ + \theta) = \cos \theta}$$

$$(b) \cos(90^\circ + \theta) = \cos 90^\circ \cos \theta - \sin 90^\circ \sin \theta$$

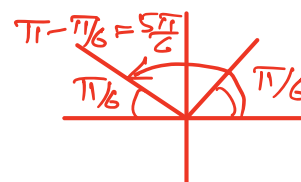
$$= (0) - (1) \sin \theta$$

$$\therefore \boxed{\cos(90^\circ + \theta) = -\sin \theta}$$

$$c \sin(180^\circ - \theta) = \sin 180^\circ \cos \theta - \cos 180^\circ \sin \theta$$

$$= 0 \cos \theta - (-1) \sin \theta$$

$$\therefore \boxed{\sin(180^\circ - \theta) = \sin \theta}$$



$$e \sin(2\pi - A) = \sin 2\pi \cos A - \cos 2\pi \sin A$$

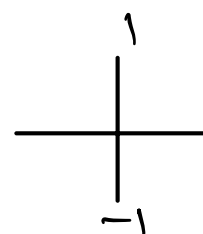
$$= 0 - (1) \sin A$$

$$\boxed{\sin(2\pi - A) = -\sin A}$$

$$f \cos(\frac{3\pi}{2} - \theta) = \cos(\frac{3\pi}{2}) \cos \theta + \sin(\frac{3\pi}{2}) \sin \theta$$

$$= 0 + (-1) \sin \theta$$

$$\therefore \boxed{\cos(\frac{3\pi}{2} - \theta) = -\sin \theta}$$



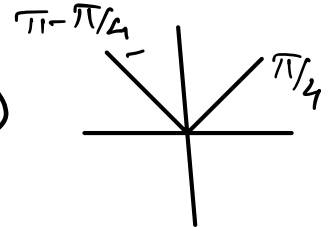
odd multiple
of $\frac{\pi}{2}$ in $\cos \theta$
is zero

$$\begin{aligned}
 \text{g } \tan\left(\frac{\pi}{4} + \theta\right) &= \frac{\tan\frac{\pi}{4} + \tan\theta}{1 - \tan\frac{\pi}{4}\tan\theta} \\
 &= \frac{1 + \tan\theta}{1 - \tan\theta}
 \end{aligned}$$



	30° $\frac{\pi}{6}$	45° $\frac{\pi}{4}$	60° $\frac{\pi}{3}$
Sin	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$
cos	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$
Tan	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$

$$\begin{aligned}
 \text{h } \tan\left(\theta - \frac{3\pi}{4}\right) &= \frac{\tan\theta - \tan\frac{3\pi}{4}}{1 + (\tan\theta)(\tan\frac{3\pi}{4})} \\
 &= \frac{\tan\theta - 1}{1 + \tan\theta(-1)} \\
 &= \frac{\tan\theta - 1}{1 - \tan\theta}
 \end{aligned}$$



4 Simplify using the compound angle identities in reverse:

1E a $\cos 2\theta \cos \theta + \sin 2\theta \sin \theta$

c $\cos A \sin B - \sin A \cos B$

e $\sin \phi \sin \theta - \cos \phi \cos \theta$

g $\frac{\tan 2\theta - \tan \theta}{1 + \tan 2\theta \tan \theta}$

b $\sin 2A \cos A + \cos 2A \sin A$

d $\sin \alpha \sin \beta + \cos \alpha \cos \beta$

f $2 \sin \alpha \cos \beta - 2 \cos \alpha \sin \beta$

h $\frac{\tan 2A + \tan A}{1 - \tan 2A \tan A}$

(a) $\cos 2\theta \cos \theta + \sin 2\theta \sin \theta = \cos(2\theta - \theta) = \cos \theta$

b $\sin 2A \cos A + \cos 2A \sin A = \sin(2A + A) = \sin 3A$

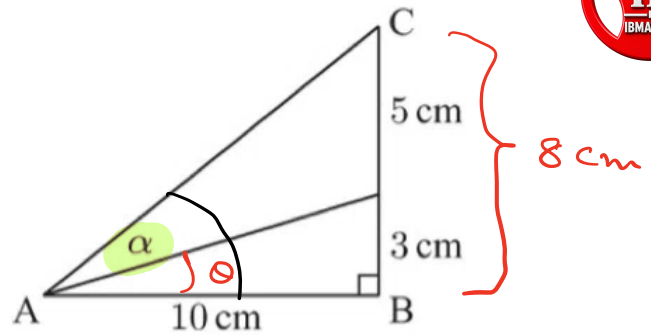
g $\frac{\tan 2\theta - \tan \theta}{1 + \tan 2\theta \tan \theta} = \tan(2\theta - \theta) = \tan \theta$

h $\frac{\tan 2A + \tan A}{1 - \tan 2A \tan A} = \tan(2A + A) = \tan 3A$

13 Find the exact value of $\tan \alpha$:

1E

$$\tan(\alpha + \theta) = \frac{8}{10}$$



$$\frac{\tan \alpha + \tan \theta}{1 - \tan \alpha \tan \theta} = \frac{8}{10}$$

$$(\tan \theta = \frac{3}{10})$$

$$\frac{\tan \alpha + (\frac{3}{10})}{1 - \tan \alpha (\frac{3}{10})} = \frac{8}{10}$$

$$\frac{10 \tan \alpha + 3}{10} = \frac{8}{10}$$

$$\frac{10 - \tan \alpha}{10}$$

$$\frac{10 \tan \alpha + 3}{10 - \tan \alpha} = \frac{8}{10}$$

$$100 \tan \alpha + 30 = 80 - 8 \tan \alpha$$

$$100 \tan \alpha + 8 \tan \alpha = 80 - 30$$

$$108 \tan \alpha = 50$$

$$\therefore \tan \alpha = \frac{50}{108} = \frac{25}{54}$$

15 Simplify:

1E a $\cos(\alpha + \beta) \cos(\alpha - \beta) - \sin(\alpha + \beta) \sin(\alpha - \beta)$

b $\sin(\theta - 2\phi) \cos(\theta + \phi) - \cos(\theta - 2\phi) \sin(\theta + \phi)$

H.W. c $\cos \alpha \cos(\beta - \alpha) - \sin \alpha \sin(\beta - \alpha)$

d $\tan(A + \frac{\pi}{4}) \tan(A - \frac{\pi}{4})$

e $\frac{\tan(A+B) + \tan(A-B)}{1 - \tan(A+B)\tan(A-B)}$ H.W.

a $\cos(\alpha + \beta) \cos(\alpha - \beta) - \sin(\alpha + \beta) \sin(\alpha - \beta)$

$= \cos((\alpha + \beta) + (\alpha - \beta)) = \cos 2\alpha$

b $\sin(\theta - 2\phi) \cos(\theta + \phi) - \cos(\theta - 2\phi) \sin(\theta + \phi)$

$= \sin((\theta - 2\phi) - (\theta + \phi)) = \sin(\theta - 2\phi - \theta - \phi)$

$= \sin(-3\phi) = -\sin 3\phi$

d $\tan(A + \frac{\pi}{4}) \tan(A - \frac{\pi}{4}) =$

$\left\{ \tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B} \right.$

$= \left[\frac{\tan A + \tan \frac{\pi}{4}}{1 - \tan A \tan \frac{\pi}{4}} \right] \left[\frac{\tan A - \tan \frac{\pi}{4}}{1 + \tan A \tan \frac{\pi}{4}} \right]$

$= \left(\frac{\tan A + 1}{1 - \tan A} \right) \left(\frac{\tan A - 1}{1 + \tan A} \right)$

$= \frac{\tan^2 A - 1}{1 - \tan^2 A}$

$((a+b)(a-b) = a^2 - b^2)$

16 If $\sin A = \frac{2}{3}$, $\frac{\pi}{2} \leq A \leq \pi$ and $\cos B = -\frac{4}{5}$, $\pi \leq B \leq \frac{3\pi}{2}$, find:

1E a $\tan(A+B)$ $\leftarrow$ b $\tan 2A$

(a) $\tan(A+B)$

$$= \frac{\sin(A+B)}{\cos(A+B)}$$

$$= \frac{\sin A \cos B + \cos A \sin B}{\cos A \cos B - \sin A \sin B}$$

$$= \frac{\left(\frac{2}{3}\right)\left(-\frac{4}{5}\right) + \left(-\frac{\sqrt{5}}{3}\right)\left(-\frac{3}{5}\right)}{\left(-\frac{\sqrt{5}}{3}\right)\left(-\frac{4}{5}\right) - \left(\frac{2}{3}\right)\left(-\frac{3}{5}\right)}$$

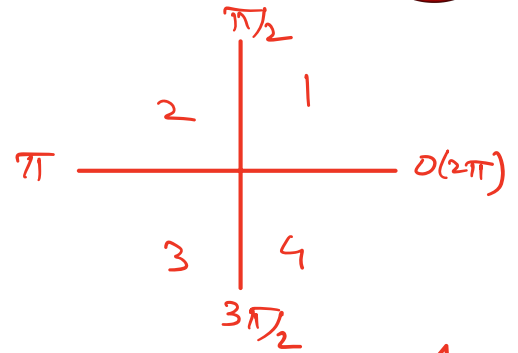
$$= \frac{-8 + 3\sqrt{5}}{4\sqrt{5} + 6}$$

$$= \frac{-8 + 3\sqrt{5}}{2(2\sqrt{5} + 3)} \times \frac{(2\sqrt{5} - 3)}{(2\sqrt{5} - 3)}$$

$$= \frac{-16\sqrt{5} + 24 + (6 \times 5) - (9\sqrt{5})}{2((2\sqrt{5})^2 - (3)^2)}$$

$$= \frac{54 - 25\sqrt{5}}{2(20 - 9)}$$

$$\tan(A+B) = \frac{54 - 25\sqrt{5}}{22}$$

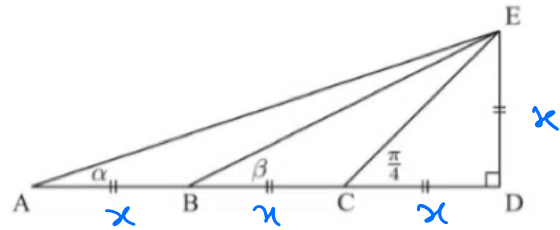


$\sin A = \frac{2}{3}$
 $x = \sqrt{9 - 4}$
 $x = \sqrt{5}$
 $\cos A = -\frac{\sqrt{5}}{3}$

$\cos B = -\frac{4}{5}$
 $y = \sqrt{25 - 16}$
 $= \sqrt{9} = 3$
 $\sin B = -\frac{3}{5}$

20 Prove that, in the given figure, $\alpha + \beta = \frac{\pi}{4}$.

IE



$$\tan \frac{\pi}{4} = \frac{x}{x} = 1$$

$$\tan \beta = \frac{x}{2x} = \frac{1}{2}$$

$$\tan \alpha = \frac{x}{3x} = \frac{1}{3}$$

$$\therefore \tan(\alpha + \beta)$$

$$= \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$= \frac{(\frac{1}{3}) + (\frac{1}{2})}{1 - (\frac{1}{3})(\frac{1}{2})}$$

$$= \frac{\frac{2+3}{6}}{\frac{6-1}{6}} = \frac{5/6}{5/6} = 1$$

$$\therefore \tan(\alpha + \beta) = 1$$

$$\therefore (\alpha + \beta) = \tan^{-1}(1) = \pi/4$$

$$\therefore \alpha + \beta = \pi/4$$

Example 16

Self Tutor

Suppose $\sin x - \sqrt{3} \cos x = k \cos(x + b)$ for $k > 0$ and $0 < b < 2\pi$. Find k and b .

$$\begin{aligned}\sin x - \sqrt{3} \cos x &= k \cos(x + b) \\ &= k[\cos x \cos b - \sin x \sin b] \\ &= k \cos x \cos b - k \sin x \sin b\end{aligned}$$

Equating coefficients of $\cos x$ and $\sin x$, $k \cos b = -\sqrt{3}$ (1) and $-k \sin b = 1$ (2)

$$\frac{-k \sin b}{k \cos b} = -\frac{1}{\sqrt{3}} \quad \{\text{dividing (2) by (1)}\}$$

$$\therefore \tan b = \frac{1}{\sqrt{3}}$$

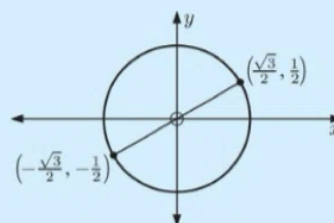
$$\therefore b = \frac{\pi}{6} \text{ or } \frac{7\pi}{6}$$

Substituting $b = \frac{\pi}{6}$ into (1) gives $k \times \frac{\sqrt{3}}{2} = -\sqrt{3}$
 $\therefore k = -2$

We reject this solution as $k > 0$.

Substituting $b = \frac{7\pi}{6}$ into (1) gives $k \times -\frac{\sqrt{3}}{2} = -\sqrt{3}$
 $\therefore k = 2$

So, $k = 2$ and $b = \frac{7\pi}{6}$.



23 Suppose $\sqrt{3} \sin x + \cos x = k \sin(x + a)$ for $k > 0$ and $0 < a < 2\pi$. Find k and a .

1E

$$\sqrt{3} \sin x + \cos x = k [\sin x \cos a + \cos x \sin a]$$

$$\therefore \sqrt{3} \sin x + \cos x = k \cos a \sin x + k \sin a \cos x$$

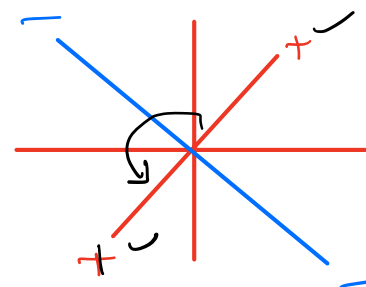
equating coeff. of $\sin x$ & $\cos x$

$$\sqrt{3} = k \cos a, \quad 1 = k \sin a.$$

$$\frac{k \sin a}{k \cos a} = \frac{1}{\sqrt{3}}$$

$$\tan a = \frac{1}{\sqrt{3}}$$

$$a = \frac{\pi}{6}, \quad a = \pi + \frac{\pi}{6} = \frac{7\pi}{6}$$



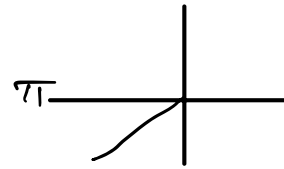
$$k \cos a = \sqrt{3}$$

$$k \cos \frac{\pi}{6} = \sqrt{3} \Rightarrow k \left(\frac{\sqrt{3}}{2}\right) = \sqrt{3} \Rightarrow \underline{k = 2}$$

$$k \cos\left(\frac{7\pi}{6}\right) = \sqrt{3}$$

$$k\left(-\frac{\sqrt{3}}{2}\right) = \sqrt{3}$$

$$k = -2 \times$$



$$\frac{7\pi}{6} = \pi + \frac{\pi}{6}$$

$$k \sin a = 1$$

$$k \sin\left(\frac{\pi}{6}\right) = 1$$

$$k\left(\frac{1}{2}\right) = 1 \Rightarrow \boxed{k=2}$$

$$\therefore k=2 \text{ \& } a = \frac{\pi}{6}, \frac{7\pi}{6}$$